3 **Flows in networks**	3.1	**Cuts and their capacity.**	**Only networks with directed arcs will be considered.**
	3.2	**Use of the labelling procedure to augment a flow to determine the maximum flow in a network.**	**The arrow in the same direction as the arc will be used to identify the amount by which the flow along that arc can be increased. The arrow in the opposite direction will be used to identify the amount by which the flow in the arc could be reduced.**
	3.3	**Use of the max–flow min–cut theorem to prove that a flow is a maximum flow.**	
	3.4	Multiple sources and sinks. Vertices with restricted capacity.	Problems may include vertices with restricted capacity.

Topics	What students need to learn:		
	Content		Guidance
3 **Flows in networks** *continued*	3.5	Determine the optimal flow rate in a network, subject to given constraints.	Problems may include both upper and lower capacities.